

Topic Modeling of Strava Google Play Reviews

MIS4078 Text Mining and Business Applications: Final Project

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1. App Selection and Research Motivation

For this project I chose Strava, one of the most widely used social fitness apps for runners and cyclists. Strava lets users record their activities via GPS, analyze their performance, and share and compare their results with a larger community of other athletes. I selected it because it has a very large and active user base, and its reviews are rich with discussion about performance tracking, motivation, features and technical reliability, which makes it well suited for topic modeling.

My research question is:

“What do Strava users talk about most, and to what extent do aspects of self-improvement and training feedback appear in their discussions?”

Analyzing this is meaningful because Strava positions itself as a tool to track and improve athletic performance. Looking at what users actually write reveals whether they experience the app that way, and what else drives their satisfaction or frustration. These insights are directly relevant for understanding user experience and for prioritizing product improvements.

Before running the analysis, I expected to uncover topics such as distance and performance tracking, GPC accuracy, motivation and the social/community side, pricing, and technical problems.

2. Data Description

The dataset consists of English-language user reviews of the Strava app (package id com.strava) collected from the Google Play Store using the google-play-scraper library, sorted from newest to oldest. The data is a snapshot taken in June 2026.

- Number of reviews collected: 112102
- Language: English (lang = “en”, country = “US”)
- Fields retained: user name and review text
- Source: Google Play Store, app id: com.strava

During preprocessing, empty review and reviews that were too short to be informative were removed, so the final modeling corpus is based on the remaining, content-rich reviews.

3. LDA Analysis

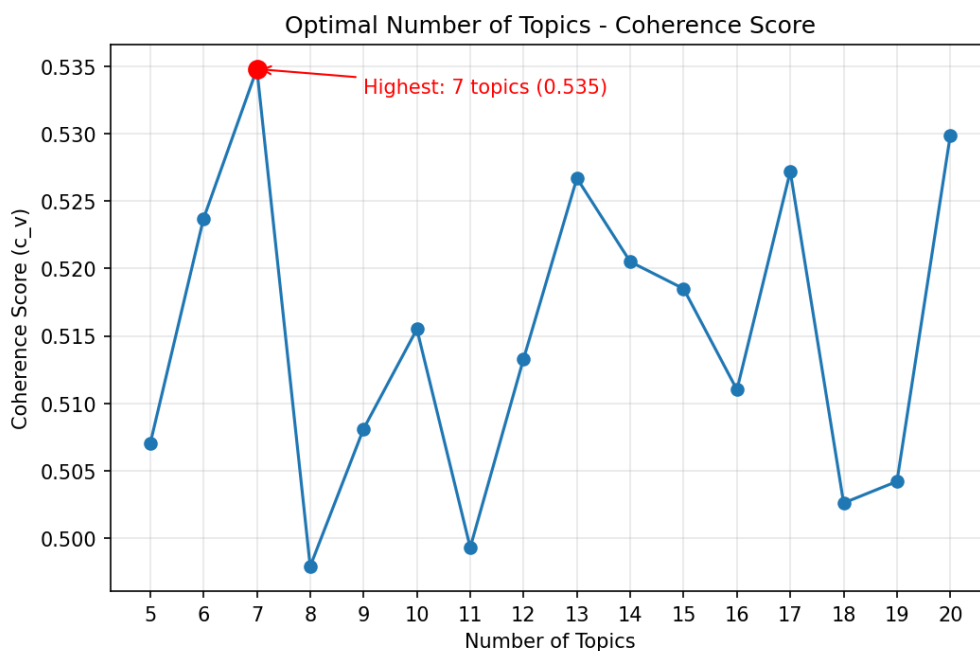
The full Python code (web scraping, preprocessing and modeling) is submitted separately as a Jupyter notebook (.ipynb). The analysis followed this pipeline:

- Text cleaning: lower-casing and removal of URLs, numbers, punctuation and extra whitespace.

- POS filtering: keeping only nouns, adjectives and meaningful verbs (removing pronouns, articles and generic verbs such as “be”, “have”, “get”), while lemmatizing each token to its base form.
- Dictionary and corpus: building a gensim dictionary, filtering extremes (*no_below* =20, *no_above*=0.9) and converting each review into a bag-of-words representation.
- Modeling: training LDA models with gensim’s LdaMulticore.

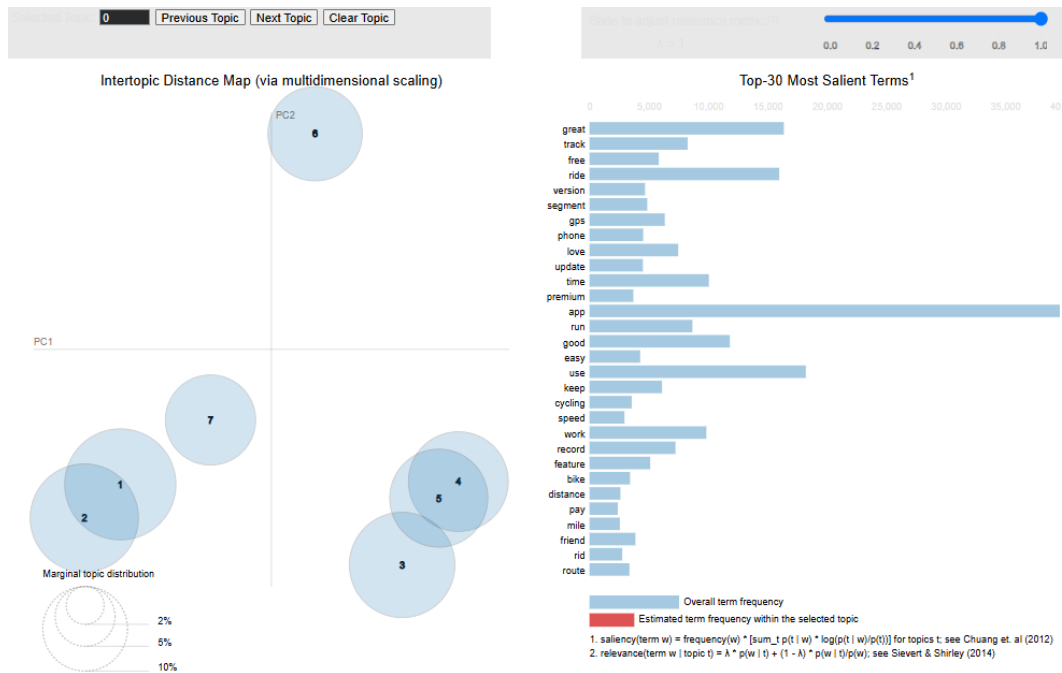
Choosing the Number of Topics

To determine the number of topics, I trained LDA models for 5 to 20 topics and computed the *c_v* coherence score for each. As shown below, coherence peaked at 7 topics (0.5348). Although a few higher topics count came close, 7 gave both the highest coherence and the most interpretable, clearly distinguishable topics, so I selected 7 topics for the final model.



Topic Model Visualization (pyLDAvis)

The final 7-topic model was visualized the pyLDAvis. The intertopic distance map below shows the topics as bubbles. Well-separated bubbles indicate clearly distinct topics, while bubble size reflects how often a topic occurs across the reviews.



Summary of Topics

The table below summarizes the seven topics, each with an assigned label, its top keywords, a representative review, and short interpretation.

Topic	Label	Top Keywords	Sample Review	Description
0	General Satisfaction & Training	App, great, good, easy, love, training, excellent, accurate, simple, motivation, cyclist	<i>"Best app going! Brilliant analysis of performance and great for reference year on year to fitness levels and training."</i>	Overall positive reviews praising Strava as a usefull, easy motivating training competition, often used together with devices such as Garmin.
1	GPS & Distance Accuracy	Gps, distance, speed, route, map, accurate, mile, signal, record, lose	<i>"Ran the same route two days in a row; Strava showed 3.44 then 3.40 miles while other apps stayed consistent."</i>	Concerns about the accuracy of GPS-based distance, speed and route measurement, with frequent reports of signal loss or wrong distances.
2	Activity Recording (Record/Upload)	Record, activity, upload, start, stop, save, pause, auto, heart, fail	<i>"The main function is recording activity, but when I press the record button, the app crashes."</i>	The core recording workflow (start, pause, save, upload) including crashes and activites that are not recorded completely.
3	Segments & Competition	Segment, compare, compete, friend, route, add, option, people, wish	<i>"Adds interest to rides. I like being able to compare rides on each segment and see what times others achieve."</i>	Comparing performance on segments and competing with friends and other athletes; also includes requests for new features.
4	Motivation & Progress (Community)	track, progress, challenge, motivate, friend, help, cycling, run, awesome, keep	<i>"Being able to share my exercise efforts with my friends motivates me to keep doing my best. Best app for keeping track of your fitness."</i>	Tracking progress, challenges and sharing with friends as a source of motivation, and the clearest expression of selfimprovement and training feedback.

Topic	Label	Top Keywords	Sample Review	Description
5	Pricing & Subscription	free, premium, pay, subscription, year, account, service, version, support	<i>"Made previously free features premium only. Making your free app worse does not make me want to give you money."</i>	Cost, the premium subscription and frustration about features that used to be free moving behind a paywall.
6	Technical Problems & Sync	update, fix, crash, android, sync, battery, issue, problem, phone, work	<i>"Since mid-January it just stopped syncing with my Fitbit and I don't know why."</i>	Bugs, crashes after updates, device-sync problems (Fitbit, Garmin, Samsung) and battery drain.

4. Discussion

Across the seven topics, two broad groups of concerns emerge. On one side, users care strongly about motivation and progress (Topic 4), general training value (Topic 0) and the competitive, social side of segments (Topic 3). These positive themes show that a large part of what users discuss is exactly the self-improvement and training-feedback dimension at the center of my research question: people value Strava for tracking their progress, being motivated by friends and challenges, and comparing their performance over time.

On the other side, a substantial share of the discussion is about problems: the accuracy of GPS and distance measurement (Topic 1), failures in the core recording workflow (Topic 2), broader technical and sync issues (Topic 6), and the cost of the premium subscription (Topic 5). These topics reveal what undermines user satisfaction.

Taken together, the answer to the research question is clear: self-improvement and training feedback are indeed prominent in what Strava users talk about, but they coexist with strong concerns about data accuracy, technical reliability and price. User satisfaction therefore appears to depend not only on Strava's motivational and social strengths, but equally on whether the recorded data can be trusted and whether the app works reliably

5. Practical Implications

If I were to present these findings to Strava's designers, developers and managers, I would make the following recommendations:

1. Prioritize GPS and recording reliability (Topics 1, 2, 6). Inaccurate distances, crashes during recording and sync failures directly damage the core value of the app, trustworthy activity data, and should be the top engineering priority.
2. Reconsider and clearly communicate the paywall (Topic 5). Moving previously free features to premium causes visible resentment. A clearer value proposition for premium, or a more flexible (e.g. monthly) option, could reduce frustration.
3. Strengthen what users already love (Topics 3, 4). Motivation, progress tracking, challenges and segments are the strongest drivers of positive sentiment. Continued investment in social features, challenges and progress feedback reinforces the app's main competitive advantage.
4. Improve device integration (Topic 6). Smoother, more stable syncing with Garmin, Fitbit and Samsung devices would remove a recurring source of complaints.

6. Conclusion

This project applied LDA topic modeling to 112102 English-language Strava reviews from the Google Play Store in order to understand what users talk about most, and whether self-improvement and training feedback are part of that conversation. The analysis identified seven clear topics, ranging from general satisfaction, motivation and progress, and segment-based competition, to GPS accuracy, recording reliability, technical and sync problems, and pricing.

The findings show that self-improvement and training feedback are indeed central to how users experience Strava: motivation, progress tracking and social comparison are among the most prominent and most positive themes. At the same time, this value is conditional. Users only benefit from these features when the underlying data is accurate, the app works reliably and the cost is perceived as fair. The strongest sources of frustration were inaccurate GPS measurements, recording failures, sync issues and the shift of previously free features behind a paywall.

For Strava, the practical message is therefore twofold: protect and reinforce the motivational and social strengths that make users love the app, while treating data accuracy, technical reliability and transparent pricing as the foundation those strengths depend on.

As a topic model, LDA describes what users discuss rather than measuring cause and effect, and this analysis is limited to English-language reviews captured at a single point in time. Future work could track how these topics shift over time, for example after major updates or pricing changes, or compare them across different languages and markets. Overall, topic modeling proved to be an effective way to turn a very large, unstructured set of reviews into a clear, corpus-level picture of user experience and satisfaction.